how to predict the noise that will plague the apparatus, and how to suppress the noise so the apparatus will succeed in its task—and from me, Braginsky was learning how to think about Einstein's laws of gravity, how to identify their predictions. We were rapidly becoming a team, each bringing to our joint enterprise his own expertise; and over the next two decades, together we would have great fun and make a few discoveries.

Each year in the early and mid-1970s, when we saw each other in Moscow or Pasadena or Copenhagen or Rome or wherever, Braginsky repeated his warning about quantum mechanical trouble for gravitational-wave detectors, and each year I again did not understand. His warning was somewhat muddled because he himself did not understand fully what was going on. However, in 1976, after Braginsky, and independently Robin Giffard at Stanford University, managed to make the warning more clear, I suddenly understood. The warning was serious, I finally realized; the ultimate sensitivity of a bar detector is severely limited by the *uncertainty principle.*

The uncertainty principle is a fundamental feature of the laws of quantum mechanics. It says that, if you make a highly accurate measurement of the position of an object, then in the process of your measurement you will necessarily kick the object, thereby perturbing the object's velocity in a random, unpredictable way. The more accurate your position measurement is, the more strongly and unpredictably you must perturb the object's velocity. No matter how clever you are in designing your measurement, you cannot circumvent this innate uncertainty. (See Box 10.2.)

Box 10.2

The Uncertainty Principle and Wave/Particle Duality

The uncertainty principle is intimately related to wave/particle duality (Box 4.1)—that is, to the propensity of particles to act sometimes like waves and sometimes like particles.

If you measure the position of a particle (or any other object, for example, the end of a bar) and learn that it is somewhere inside some error box, then regardless of what the particle's wave might have looked like